GHAZI UNIVERSITY

Department of Computer Science

LAB 3

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Programming Fundamental Lab 3

Task 1: Assume that the following variables are defined:

```
int age;
double pay;
char section;
```

Write a Single Cin statment that will read input into each of these variables.

Input

```
#include <iostream>
using namespace std;
int main ( ) {
  int age = 20;
double pay= 4500.75;
Char Section = "A";
Cout << "Page: "<<age << endl;
Cout << "Pay": << pay<< endl;
Cout << "Section:" << Section<<endl;
Cout << "-------" << endl;
}
```

output:

```
Age: 20
Pay: 4500.75
Section : A
```

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```
1
2  #include <iostream>
3  using namespace std;
4
5  int main() {
6      // Task 1 - Values put kar diye hain
7      int age = 20;
8      double pay = 4500.75;
9      char section = 'A';
10
11     cout << "Age: " << age << endl;
12     cout << "Pay: " << pay << endl;
13     cout << "Section: " << section << endl;
14     cout << "------------------" << endl;
15 }
```

TAB

```
Age: 20
Pay: 4500.75
Section: A
------------------

[Program finished]
```

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Task 2:

Complete the following table by writing the value of each expression in the value column according to C++ language rules.

Expression	Values
28/4-2	5
6+12*2-8	22
4+8*2	20
6+17%3-2	6
2+22*(9-7)	46
(8+7)*2	30
(16+7)%2-1	0
12/(10-6)	3
(19-3)*(2+2)\4	16

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Task 3: Assume a program has the following variable difinitions:

```
int units;
float mass;
Weight = mass* units;
```

which automatic data type Conversion will take place?

A. mass is demoted to an int, units remain as int, and the result of mass* units in an iht.

B. unit is promoted to a float, mass remains a float, and the result of mass* unit is float.

C. Unit is promoted to a float, mass remains a float, and the result of mass* units is a double

Input

```
#include <iostream>
using namespace std;
int main() {
int units = 5;
float mass = 2.5f;
double weight;
weight = mass* units;
Cout << "unit (int):" <<units<< endl;
Cout << " mass (float)." << mass << endl;
Cout<< "weight (mass* units) double me:" << weight << endl;
  return 0;
}
```

output:

```
b= 2
c= 4.2
b*c = 8.4
a me store hua =8
```

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```cpp
#include <iostream>
using namespace std;

int main(){
    int a, b = 2;
    float c = 4.2;

    a = b * c; // 8.4 float se 8 int ban jayega

    cout << "b = " << b << endl;
    cout << "c = " << c << endl;
    cout << "b * c = " << b * c << endl;
    cout << "a me store hua = " << a << endl;

    return 0;
}
```

TAB

```
b = 2
c = 4.2
b * c = 8.4
a me store hua = 8

[Program finished]
```

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Task 4: Assume a Program has the following Variable definitions:

int a,b = 2; float c = 4.2;

and the following Statement: a = b*c;

what values will be stored in a?

A: 8.4 **B: 8** **C: 0** **D: None of above**

Input:

Correct answer is 3.8

C++ Code:

```
# include <iostream>
using namespace std;
int main ()
{
int a,b = 2;
Float c = 4.2;
a = b* c;
Cout << "b= " << b << endl;
Cout << " c=" <<<c <<endl;
Cout << "b* c= " << b* c <<endl;
Cout<<" a me Store hua :"
return 0;
}
```

output:

b= 2

c= 4.2

b*c = 8.4

a me store hua = 8

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```cpp
#include <iostream>
using namespace std;

int main(){
    int a, b = 2;
    float c = 4.2;

    a = b * c; // 8.4 float se 8 int ban jayega

    cout << "b = " << b << endl;
    cout << "c = " << c << endl;
    cout << "b * c = " << b * c << endl;
    cout << "a me store hua = " << a << endl;

    return 0;
}
```

TAB

```
b = 2
c = 4.2
b * c = 8.4
a me store hua = 8

[Program finished]
```

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Task 5:

Input:

```
#include <iostream>
  using namespace std;
int main () {
  int qty = 7;
  int sales Reps = 2;
double units Each;
units Each = qty / sales reps;
Cout<<" without Casting:" << units each<< endl;
 Units Each = static - Cast < double > (qty) /
     Sales Reps;
Cout<< " with Casting :" <<units Each << endl;
   // output 3.5
     return 0;
  }
```

Output:

without casting : 3

with casting : 3.5

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new*

```cpp
#include <iostream>
using namespace std;

int main(){
    int qty = 7;
    int salesReps = 2;
    double unitsEach;

    // Pehle (galat) - Integer division
    unitsEach = qty / salesReps;
    cout << "Without casting: " << unitsEach << endl; // Output: 3

    // Ab Sahi - Type cast ke sath
    unitsEach = static_cast<double>(qty) / salesReps;
    cout << "With casting: " << unitsEach << endl; // Output: 3.5

    return 0;
}
```

TAB

```
Without casting: 3
With casting: 3.5

[Program finished]
```

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Task 6:

Input:

```
#include <iostream>
 using namespace std;
int main() {
int num1 = rand()% 900+100;
 int num 2 = rand()% 900 + 100;
 int answer = num 1 + num 2;
Cout << "Solve this addition Problem: \n\n";
 Cout <<"  " << num 1 << endl;
 Cout <<"+" num 2 << endl;
Cout <<"-----"<<endl;
 Cin . get ( );
Cout <<"\ncorrect Solution: \n\n";
 Cout <<"  " <<num1 << endl;
Cout <<"+ " <<num 2<< endl;
 cout <<"----- "<<endl;
 Cout <<"  " <<answer<< endl;
   return 0
}
```

output:

```
  555
+ 978
-----
```

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new*

```cpp
#include <iostream>
#include <cstdlib>  // rand() ke liye
#include <ctime>    // time() ke liye
using namespace std;

int main() {
    // Random number generator ko seed karna
    srand(time(0));

    // 100 se 999 tak 2 random numbers
    int num1 = rand() % 900 + 100;  // 100-999
    int num2 = rand() % 900 + 100;  // 100-999

    int answer = num1 + num2;

    // Problem display karna
    cout << "Solve this addition problem:\n\n";
    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----" << endl;

    // Pause - Student ke solve karne tak rukna
    cout << "\nPress ENTER when you are ready
to check the answer...";
    cin.get();  // Enter press ka wait karega

    // Correct solution display karna
    cout << "\nCorrect Solution:\n\n";
    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----" << endl;
    cout << "  " << answer << endl;

    return 0;
}
```

TAB

```
Solve this addition problem:

  555
+ 978
-------

Press ENTER when you are ready to check the answer...
```

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Home Task

Task 1:

Input:

```
# include <iostream>
using namespace std;
 int main()
 {
double number 1 = 15.5;
double number 2 = 10.2;
double Sum;
Sum = number 1 + number 2;
Cout << "first number" << number 1 << endl;
Cout << "Second Number:" << number 2 << endl;
Cout << "The sum of the two numbers is" << sum << endl;
  return 0;
 }
```

output:

First number: 15.5

Second number: 10.2

The Sum of the two numbers is 26.7

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```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1 = 15.5;
    double number2 = 10.2;
    double sum;

    sum = number1 + number2;  // Yahan add ho raha hai

    cout << "First Number: " << number1 << endl;
    cout << "Second Number: " << number2 << endl;
    cout << "The sum of the two numbers is " << sum << endl;

    return 0;
}
```

TAB

```
First Number: 15.5
Second Number: 10.2
The sum of the two numbers is 25.7

[Program finished]
```

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Task 2.

Input Code:

```
#include <iostream>
using namespace std;
int main()
{
int number1 = 10;
int number2 = 4;
float quotient;
cout << " Enter two numbers and I will devicde \n";
Cout << "the first by the second for you. \n";
Cout << " you entered: " << number1 << "and" << number2 << endl;
quotient = static_cast <float>(number1) \number2;
Cout << " Quotient is: " << quotient << endl;
return 0;
}.
```

output:

```
Enter two numbers and I will divide
the first by the Second for you.
you entered: 10 and 4
Quotient is: 2.5.
```

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```
3
4  int main()
5  {
6      int number1 = 10;
7      int number2 = 4;
8      float quotient;
9
10     cout << "Enter two numbers and I will
   divide\n";
11     cout << "the first by the second for you.\n";
12
13     cout << "You entered: " << number1 << " and
   " << number2 << endl;
14
15     quotient = static_cast<float>(number1) /
   number2;
16
17     cout << "Quotient is: " << quotient << endl;
18
19     return 0;
20 }
```

TAB

```
Enter two numbers and I will divide
the first by the second for you.
You entered: 10 and 4
Quotient is: 2.5

[Program finished]
```

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